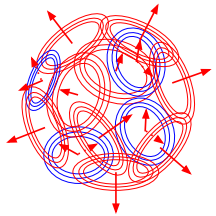
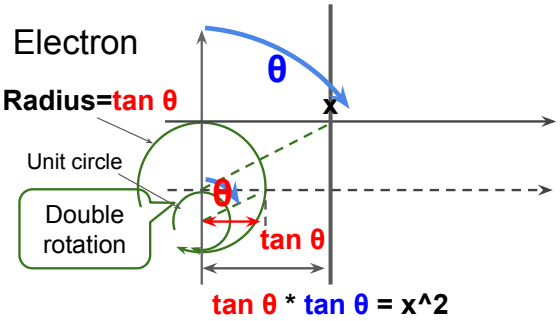
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.6: Waves and Mass



1. Mass and Energy

The famous equation “*E* = *mc*^*2*”, which states that mass is equivalent to energy, was published by Einstein in his 1905 paper, "Does the Inertia of a Body Depend Upon Its Energy-Content?".

Einstein derived this equation from a “thought experiment involving the emission of light”, which is why the speed of light is included in the formula.

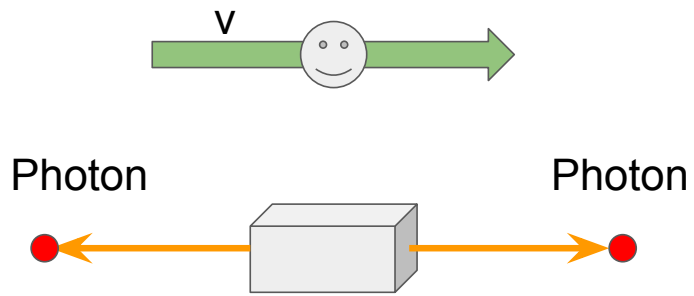
In this experiment, a stationary object is observed by an observer moving at a certain velocity relative to it.

From the observer's perspective, the object has kinetic energy because it possesses relative velocity.

1. Mass and Energy

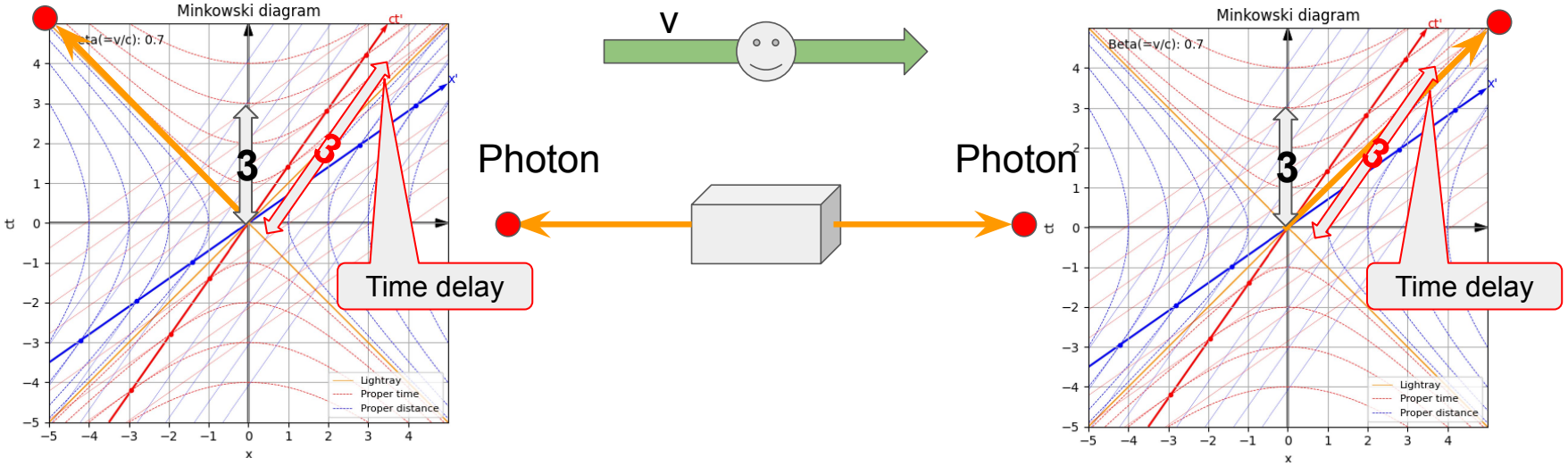
The object then emits photons both forward and backward relative to the direction of motion.

Since photons are emitted in both directions, their kinetic energy cancels out, and the total kinetic energy of the system (object plus photons) remains unchanged.



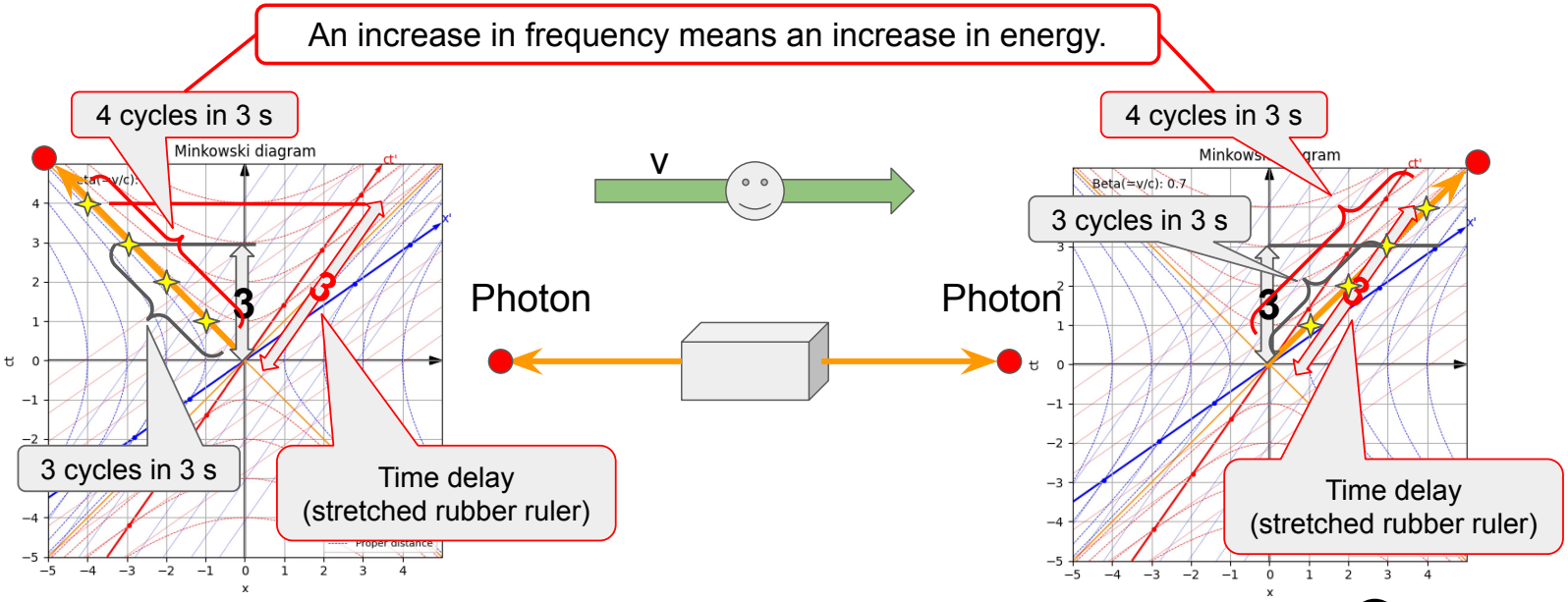
1. Mass and Energy

However, from the viewpoint of the moving observer, the energy of the photons appears to increase. This is because, according to the theory of relativity, “time appears to slow down” for a moving object.



1. Mass and Energy

Using this slowed time as a reference, the frequency of the light appears higher. This is analogous to measuring dimensions with a stretched rubber ruler; the same length (time interval) is measured as shorter.



1. Mass and Energy

A shorter time interval for light's vibration means a higher frequency, which results in the photon appearing to have “increased energy” due to time dilation (the Doppler effect in the spatial direction is canceled out because light is emitted in both directions).

According to the law of conservation of energy, the total energy of the object and the photons must be conserved even when viewed by a moving observer.

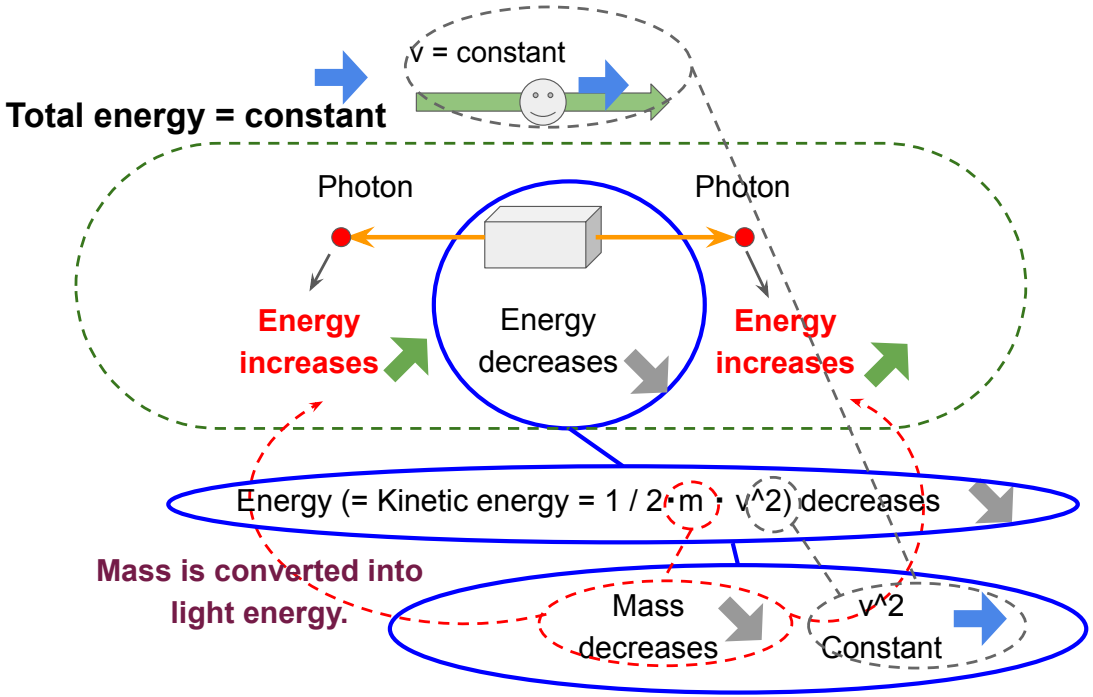
If the photon energy has increased while the total remains constant, the object's energy must decrease (or appear to decrease).

Since the object's kinetic energy is “*1/2mv^2*” and the relative velocity remains constant in this experiment, the only way for the kinetic energy to decrease is for the “mass to decrease” .

1. Mass and Energy

Therefore, the object's mass is reduced by an amount corresponding to the increase in photon energy.

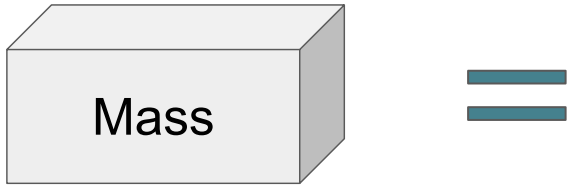
This demonstrates that mass and energy are equivalent and explains why the speed of light is included in “*E* = *mc*^2”.

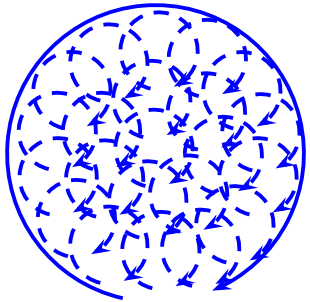


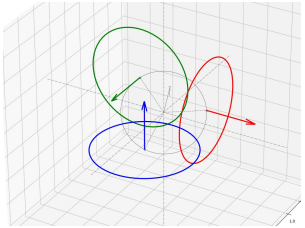
1. Mass and Energy

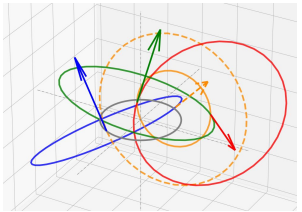
In R-space Theory, photons, electrons, and all elementary particles are composed of ”vortices of angular momentum”, and mass itself is a manifestation of this angular momentum.

Vortices of angular momentum









2. The Schrödinger Equation and Mass

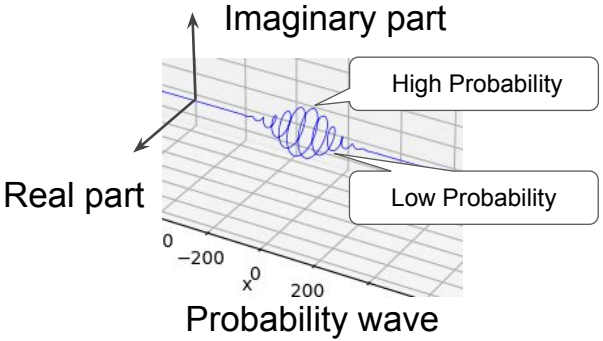
The Schrödinger equation, or wave function, represents the shape of the wave (matter wave) possessed by a particle and was published by Schrödinger in 1926. Born's later probabilistic interpretation clarified that the magnitude of this complex probability wave represents the probability of observing the quantum.

Wave function

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\psi(x,t)$$

Probability density

$$P(x,\ t) = |\psi(x,\ t)|^2$$



2. The Schrödinger Equation and Mass

While Schrödinger derived the wave function through an analogy with optics and Hamiltonian mechanics (the Hamilton-Jacobi equation), it can also be derived from de Broglie's matter waves using modern methods.

By taking the spatial and temporal derivatives of a plane wave corresponding to a free particle and comparing them with de Broglie's and the Planck-Einstein relations, one can see that the differential operators correspond to momentum and energy.

Plane wave

$$\psi(x,t) = Ae^{i(kx-\omega t)}$$

$$\frac{\partial \psi}{\partial x} = \frac{\partial}{\partial x} Ae^{i(kx-\omega t)} = ikAe^{i(kx-\omega t)} = ik\psi$$

$$\frac{\partial \psi}{\partial t} = \frac{\partial}{\partial t} Ae^{i(kx-\omega t)} = -i\omega Ae^{i(kx-\omega t)} = -i\omega\psi$$

2. The Schrödinger Equation and Mass

Plane wave

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$$\frac{\partial \psi}{\partial x} = ik\psi$$

$$\frac{\partial \psi}{\partial t} = -i\omega\psi$$

Multiply both sides by $i\hbar$.

&

$$i\hbar\frac{\partial \psi}{\partial x} = -p\psi \qquad \hat{p} = -i\hbar\frac{\partial}{\partial x}$$

$$i\hbar\frac{\partial \psi}{\partial t} = E\psi \qquad \hat{E} = i\hbar\frac{\partial}{\partial t}$$

De Broglie relation

$$p = \frac{h}{\lambda} = \hbar k$$

Planck–Einstein relation

$$E = \hbar\omega$$

2. The Schrödinger Equation and Mass

In classical mechanics, the energy of a one-dimensional particle is given by ”*E = p^2/2m + V(x)*”.

Kinetic energy Potential energy

$$E = \frac{p^2}{2m} + V(x)$$

Substituting the differential operators for “*p*” and “*E*” into this equation yields the Schrödinger equation.

The energy of a one-dimensional classical particle

$$\hat{E} = i\hbar\frac{\partial}{\partial t} \qquad E = \frac{p^2}{2m} + V(x) \qquad \hat{p} = -i\hbar\frac{\partial}{\partial x}$$

Wave function

$$i\hbar\frac{\partial}{\partial t}\psi(x,t) = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\psi(x,t)$$

2. The Schrödinger Equation and Mass

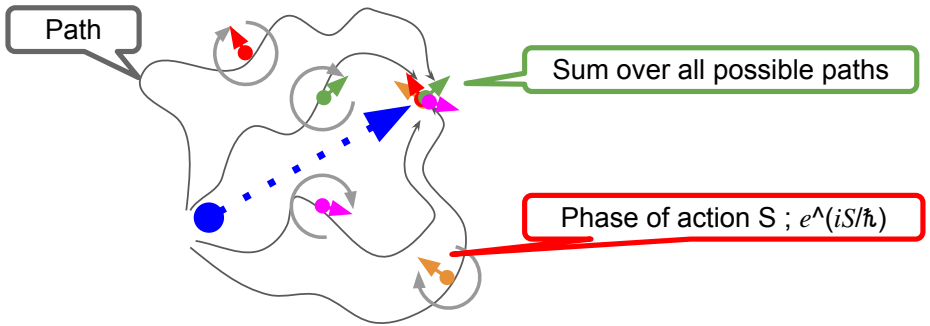
This shows that “the mass in the Schrödinger equation originates from the kinetic energy of a one-dimensional particle in classical mechanics.

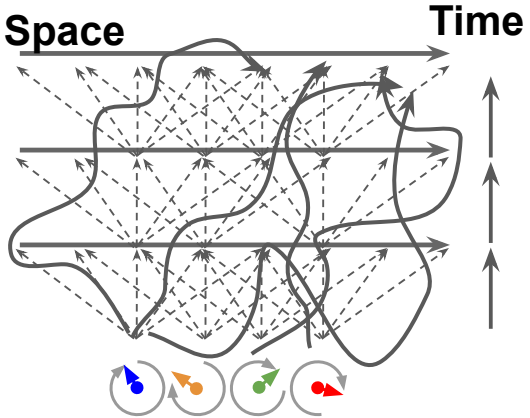
In R-space Theory, squaring “$\hbar$” or the momentum operator (operating twice) corresponds to “double rotation”.

3. Path Integrals and Mass

Path integration is a concept introduced by Feynman in 1948 in "Space-Time Approach to Non-Relativistic Quantum Mechanics," where quantum mechanics is expressed as the “sum of all possible paths”.

Each path's contribution is weighted by the phase of the action “*S*”, “e^iS/ℏ”.



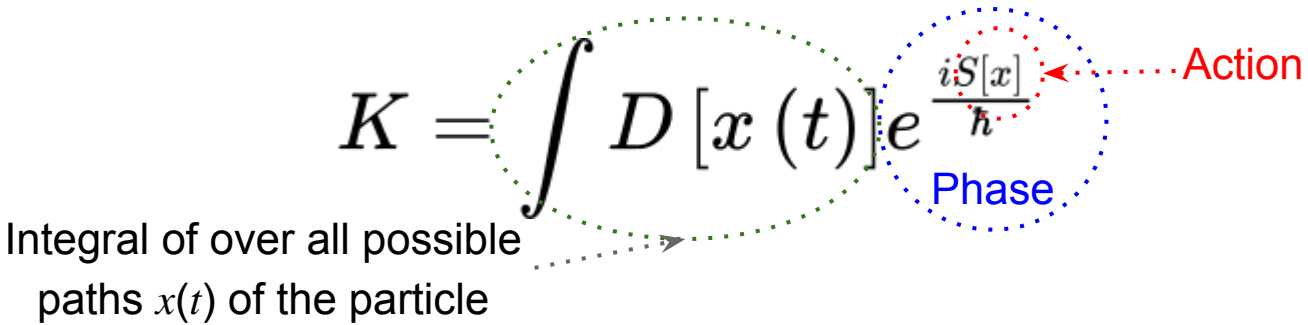


3. Path Integrals and Mass

The propagation kernel, representing the integral over all paths, satisfies the Schrödinger equation.

A larger action results in a faster-rotating phase, and paths with similar actions interfere constructively as their phases align.

Propagation kernel



3. Path Integrals and Mass

Action "*S*" is the integral of the difference between kinetic energy "*T*" and potential energy "*V*".

Kinetic energy

Potential energy = 0

$$S = \int (T - V)\, dt \quad \blacktriangleright \quad S[x] = \int_o^t \frac{m}{2}\dot{x}^2\, dt = \int_o^t \frac{m}{2} v^2 dt$$

For a free particle (with zero potential energy), the action is given by the kinetic energy "*1/2mv^2*". Using this, the propagator can be calculated and takes the following form.

As in the Schrödinger equation, we see that the mass originates from the kinetic energy of a one-dimensional classical particle.

$$K(x, T;\ x_0, 0) = \sqrt{\frac{m}{2\pi i\hbar T}} \exp\left(\frac{im(x - x_o)^2}{2\hbar T}\right)$$

$$T = t_f - t_i$$

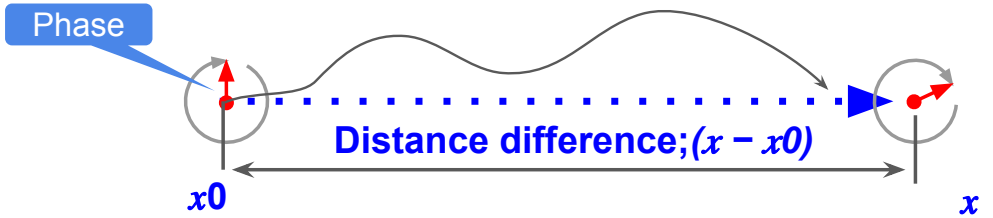
Note; "T is the difference between the initial and final times, that is, the elapsed time."

3. Path Integrals and Mass

In the propagation kernel for a free particle, the phase varies in proportion to the “square of the distance difference; *(x - x0)^2*”.

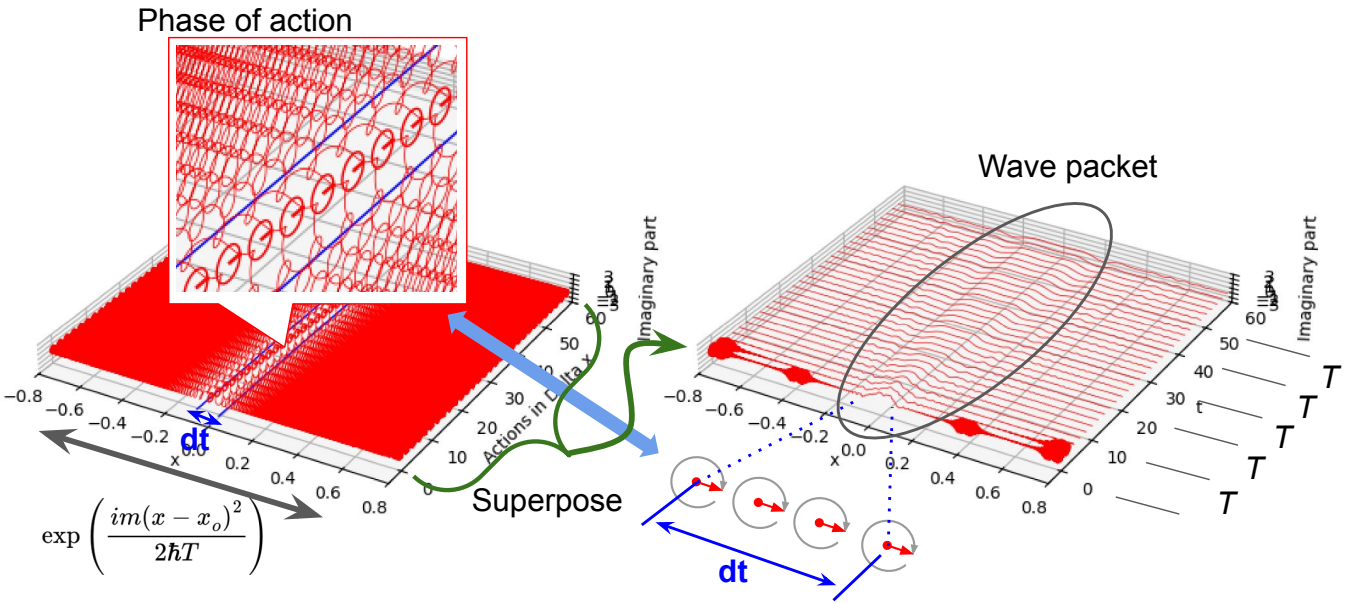
The phase varies in proportion to the squared distance difference ***(x − x0)^2***.

$$K\left(x,T;\ x_0,0\right)=\sqrt{\frac{m}{2\pi i\hbar T}}\exp\left(\frac{im\left(x-x_o\right)^2}{2\hbar T}\right)$$



3. Path Integrals and Mass

Because of this squared distance relationship, phases become misaligned as the distance increases, leading to cancellation and the formation of a “wave packet”.

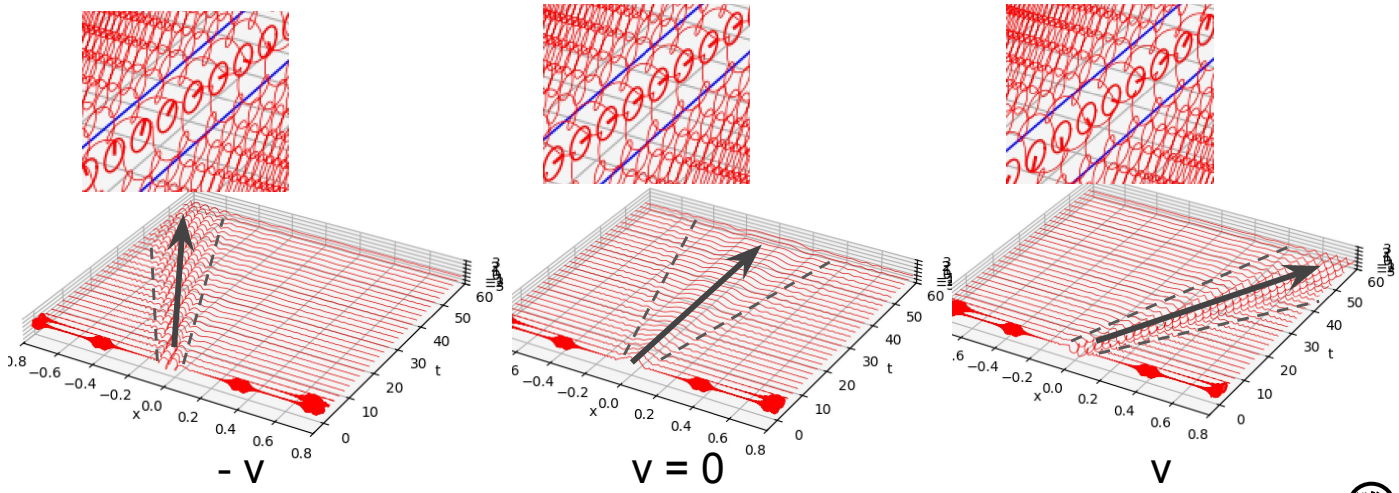


3. Path Integrals and Mass

The magnitude of this wave packet indicates the probability of observation.

The position of the peak of this wave packet remains stationary when the initial phases are aligned, but it moves when the phases vary periodically.

The peak of the wave packet represents the average velocity of the quantum.

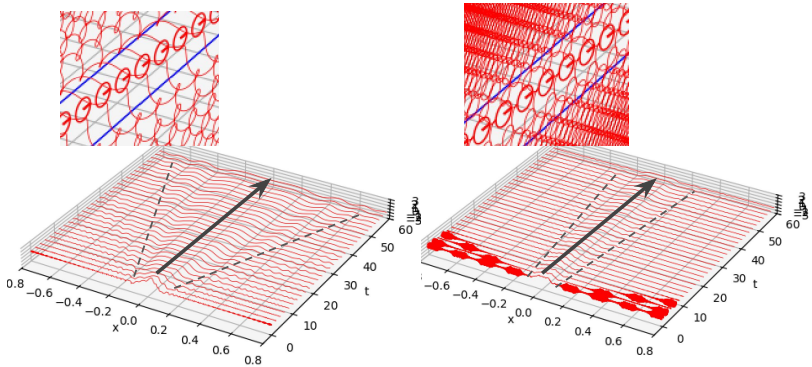


3. Path Integrals and Mass

Since mass (*m*) is a coefficient of the “*(x - x0)^2*” term, “a larger mass leads to more intense phase changes” (more rotations).

When phase changes rapidly with distance, it becomes harder for the superimposed phases to align, causing them to cancel out .

This manifests as the characteristic of mass where a "heavy" quantum is “less likely to move far”.



Small Mass Large Mass

3. Path Integrals and Mass

Furthermore, the units of the term “*m(x - x0)^2/t*” in the exponent are “kg m2/s”, which are the same as “angular momentum”.

Thus, the phase change of a quantum state depends on the ratio of the quantum's angular momentum to the Planck constant (the minimum unit of angular momentum), with “mass representing that ratio”.

$$K\left(x,T;\ x_0,0\right)=\sqrt{\frac{m}{2\pi i\hbar T}}\exp\left(\frac{im\left(x-x_o\right)^2}{2\hbar T}\right)$$

Unit: [ħ] = [ħ] = J · s = kg·m^2/s
= Unit of angular momentum ($L=r\cdot p$)
Dimension: [ħ] = $ML^{2}T$^−1

=

Unit: [m(x − x0)^2 / T]
= kg·m^2/s
Dimension: [m(x − x0)^2 / T]
= $ML^{2}T$^−1

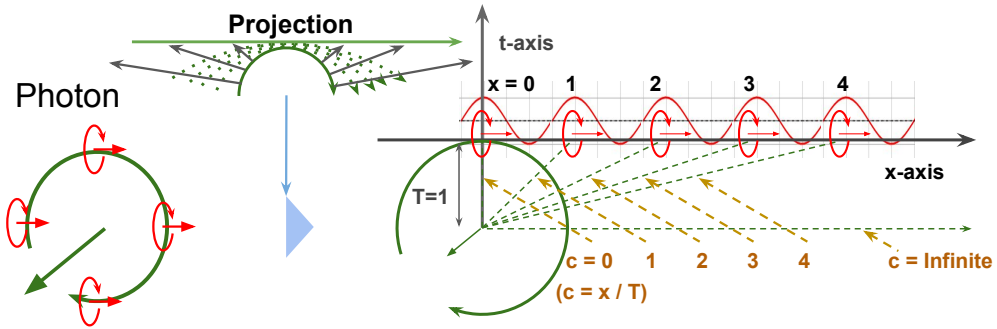
M: Mass, L: Length, T: Time

4. R-space Theory and Mass

R-space Theory uses phase changes based on action—similar to the Schrödinger equation and Feynman's path integrals—as its foundation, with the spreading of phase in R-space being inspired by path integrals.

In R-space, the phase of a photon's rotation vector (angular momentum) is projected into space.

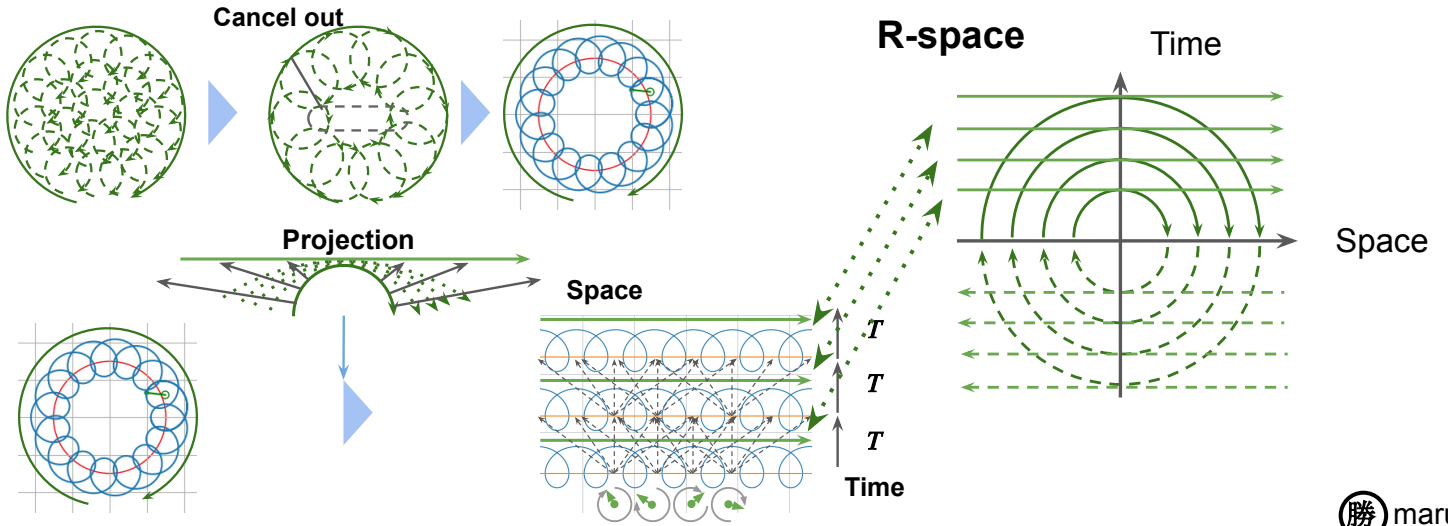
Since the angle (tilt) of the phase is perceived as velocity in projected space, each phase spreads throughout the entire space simultaneously.



4. R-space Theory and Mass

The photon's angular momentum is projected somewhere within this spreading phase, serving as the propagation kernel.

As rotation vectors are collections of small vortices, their phases are projected and evolve over time through superposition, corresponding to path integration.



4. R-space Theory and Mass

If we consider the phase circle as a unit circle (radius = 1), the projected position on the spatial axis is “$x = \tan \theta$”.

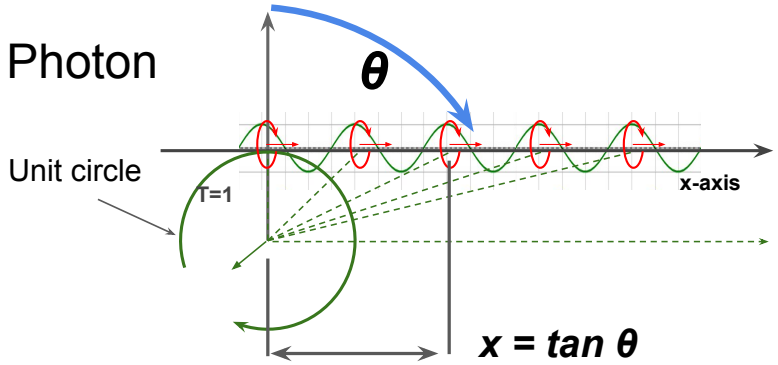
As explained in Chapter 3, an electron's rotation vector undergoes “double rotation” and possesses “1/2 rotational symmetry”.

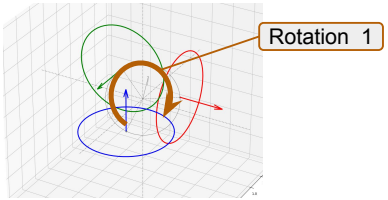
Double rotation means rotating a system that is already rotating.

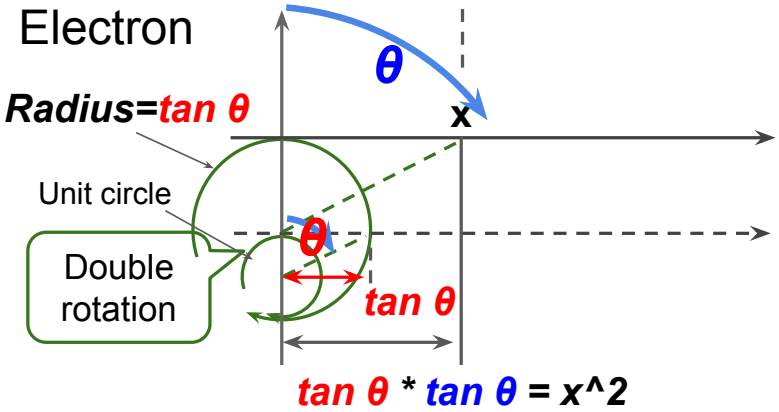
The projection of the first rotation is “$\tan \theta$”, while the second rotation—using the first projection as its radius—results in “$\tan \theta * \tan \theta = x^2$”.

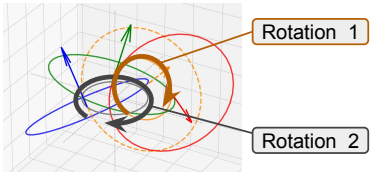
Because observers only perceive the velocity resulting from the double rotation, the phase projection of an electron's rotation vector appears proportional to the “square of the distance (x^2)”.

4. R-space Theory and Mass









5. Action, Rotation, and Mass

The phrase “rotation and phase in R-space” can be regarded simply as a graphical representation of the propagator in path analysis.

Why does this interpretation arise?

Action can be expressed in Hamiltonian form as “*S* = *px* - *Et*”.

$$S = \int L\, dt \quad \blacktriangleright \quad S = \int (p\dot{q} - H)dt$$

Hamiltonian $\Longleftrightarrow$ Lagrangian

$H(q,\ p,\ t) = p\dot{q} - L$ (Legendre transform)

$L = p\dot{q} - H$

$$\blacktriangleright \quad S = \int (p\, dq - H\, dt)$$

$dq = \dot{q}\, dt \quad p\dot{q}\, dt = p\, dq$

5. Action, Rotation, and Mass

Comparing this with a plane wave equation reveals the same relationship as applying de Broglie and Planck-Einstein relations to a wave.

$$S = \int (p\dot{q} - H)dt \quad \blacktriangleright \quad S = \int \left(p\,dx - \frac{p^2}{2m}dt \right)$$

$$p = mv$$

$$H = pv - \frac{1}{2}mv^2 = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

Action

$$S = px - Et$$

De Broglie relation

$$p = \frac{h}{\lambda} = \hbar k$$

Plane wave

$$\psi(x,t) = Ae^{i(kx-\omega t)}$$

Planck–Einstein relation

$$E = \hbar\omega$$

5. Action, Rotation, and Mass

R-space Theory diagrams represent rotation and phase, making them identical to wave equations.

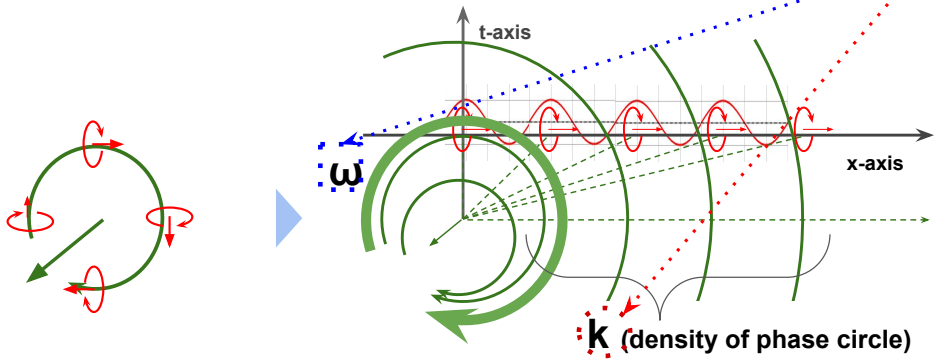
By treating this rotation as angular momentum rather than mere rotation, it represents action.

Action

$$S = px - Et$$

Plane wave

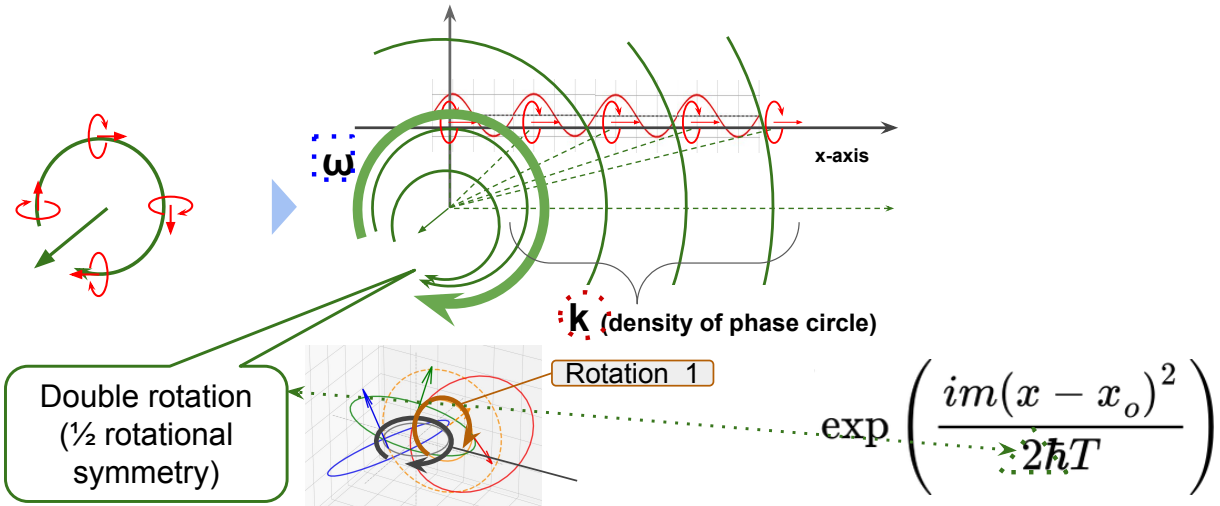
$$\psi(x,t) = Ae^{i(kx-\omega t)}$$



5. Action, Rotation, and Mass

Projecting R-space into space is equivalent to spatial differentiation.

Furthermore, the “2” in the denominator of the exponent of the propagator appears because the electron possesses a half-turn rotational symmetry; it converts the full 720° rotation into an effective 360° rotation.



5. Action, Rotation, and Mass

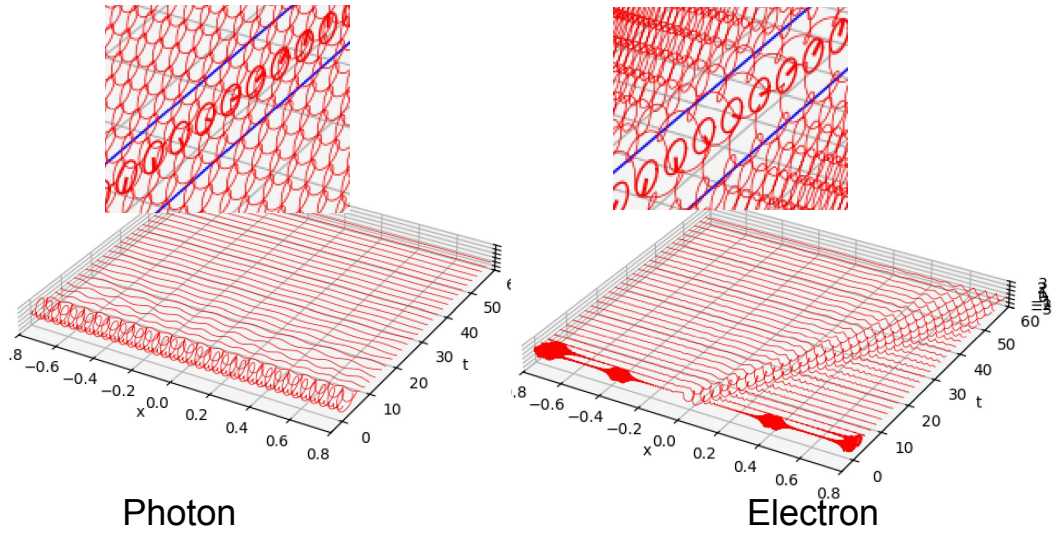
Mass is essentially a “ratio (or count) of how many minimum units of angular momentum a quantum possesses”, representing the magnitude of the quantum's vortex rotational energy.

In R-space Theory, all quanta are combinations of rotation vectors (angular momentum), which is energy itself; therefore, “mass and energy are equivalent”.

While photons also possess angular momentum “$E=hv$”), they have zero mass because their rotation vectors do not undergo double rotation.

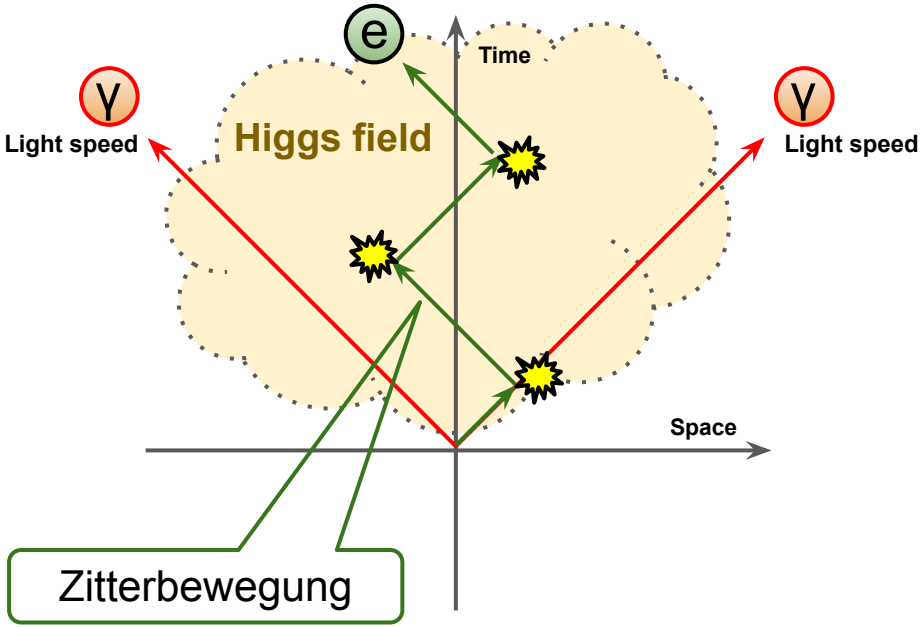
5. Action, Rotation, and Mass

Consequently, photon phases are projected proportional to distance (to the first power), and they do not form wave packets through phase superposition like electrons do.



5. Action, Rotation, and Mass

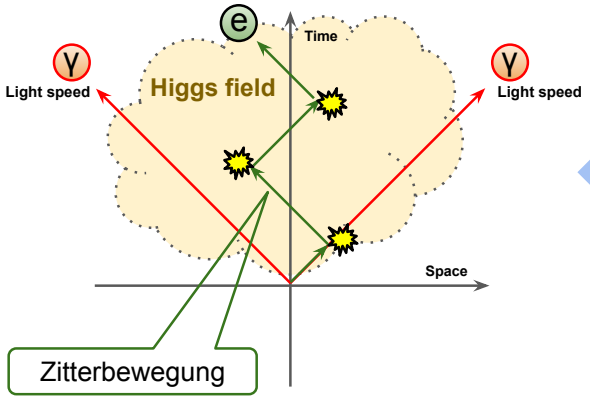
In the Standard Model, mass is explained by the interaction with the “Higgs field”, which causes a high-speed back-and-forth motion called “Zitterbewegung” (jittery motion) derived from the Dirac equation.

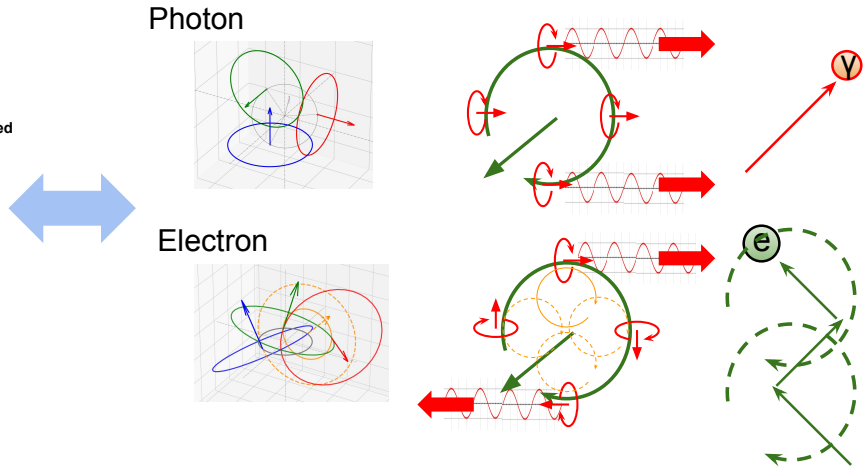


5. Action, Rotation, and Mass

In R-space Theory, this Zitterbewegung arises from the “mutual influence of the electron's three rotation vectors (x, y, z)”, causing the direction of the rotation vectors themselves to rotate.

A larger angular momentum leads to more frequent reciprocating motion, thereby increasing the mass.





5. Action, Rotation, and Mass

Moreover, the idea that the electron's Zitterbewegung can be explained by a circulating internal photon has already been proposed in the WvdM electron model introduced in the previous chapter.

As will be explained in later chapters, quanta other than the electron that possess mass are also combinations of rotational vectors, so the mechanism by which mass arises is fundamentally the same — everything originates from angular momentum.

The relationship between the medium that carries angular momentum (the R-aether) and the Higgs field will also be discussed in subsequent chapters.

Ultimately, "mass is the magnitude of the unevenness in angular momentum waves caused by 1/2 spin".

5. Action, Rotation, and Mass

References:

- *The Quantum Universe (and Why Anything That Can Happen, Does) Copyright (c) 2011 by Brian Cox and Jeff Forshaw*
- *Kondo koso wakaru (Now I Finally Understand) Feynman's Path Integrals Copyright (c) 2015 by Sumio Wada*

This is all for my presentation.

Thanks for watching!

